

Phase 1 — Wind Speed: Statistical Properties

Analysis of Results & Literature Comparison

Quantitative Energy Research
Mathematical Finance RAship
Proxy Dataset: `data.csv`

Sample period: January 2015 – December 2018

Based on: Šaltytė Benth & Benth (2008) · Tuller & Brett (1984) · Garcia et al. (1998) ·
Alexandrinis (2013)

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Overview and Purpose of Phase 1

Phase 1 is the *preprocessing* phase of the entire modelling pipeline. Its goal is not yet to forecast wind speed or price derivatives: it is to answer a more fundamental question — statistically speaking, what kind of object is daily average wind speed? — and then to transform the raw series into something that the subsequent time-series machinery can handle.

To understand why this matters, consider what the full model eventually needs to do. The wind futures pricing formula derived in Šaltytė Benth & Benth (2008) requires the *conditional distribution* of future wind speed to be Gaussian — specifically, it requires $X_1(s) \mid \mathcal{F}_t$ to be normally distributed so that the expected payoff of a futures contract can be written in closed form. But raw wind speed is never Gaussian: it is bounded below by zero, right-skewed, heavy-tailed, and seasonally non-stationary. Phase 1 is the systematic removal of all these properties through a sequence of five operations:

1. Documenting the non-normality of raw wind speed (descriptive statistics).
2. Characterising the marginal distribution (Weibull vs. lognormal).
3. Applying the Box-Cox power transformation to Gaussianise the series.
4. Removing the seasonal mean $\Lambda(t)$ via a truncated Fourier series.
5. Removing the seasonal variance $\sigma^2(t)$ via a second Fourier fit.

The output of this chain is the *standardised series* $\tilde{W}(t)$, which is zero-mean, unit-variance, and approximately white noise. It is this series — and only this series — that enters the AR(4)/CAR(4) framework in Phases 2 and 4.

Dataset Description (Cell 1)

The raw dataset (`data.csv`) contains 35,064 hourly wind speed observations measured at a meteorological station at approximately 10 m above ground level, spanning from 31 December 2014 to 31 December 2018. The data are complete: there are zero missing values. After aggregating to daily averages by arithmetic mean of the 24 hourly readings within each calendar day, we obtain $n = 1,462$ daily average wind speed (DAWS) observations covering approximately four complete calendar years.

Literature context. Šaltytė Benth & Benth (2008) use 7,550 daily observations spanning roughly 20 years, recorded at a U.S. meteorological station. Alexandrinis (2013) similarly uses 20 years of NOAA data from New York. Our proxy dataset is therefore considerably shorter (4 years vs. 20), which has direct consequences for parameter estimation uncertainty — particularly for the seasonal variance function, where only ≈ 4 residual observations are available per calendar day.

One important point about measurement height: the data are recorded at ≈ 10 m, whereas operational wind turbines have hub heights of 80–120 m. Under the standard power-law vertical wind profile:

$$V(z) = V_{10} \cdot \left(\frac{z}{10}\right)^\alpha, \quad \alpha \approx 0.14 \text{ (open terrain)}, \quad (1)$$

wind speed at 100 m is roughly $100^{0.14}/10^{0.14} \approx 1.74$ times the 10 m reading. This correction is not applied in Phase 1 because the dataset is used only to validate the preprocessing

pipeline; the physically representative hub-height values come in Phase 2. The reader should therefore interpret all absolute speed values in this document as proxy figures, not as wind-resource assessments.

Descriptive Statistics (Cell 2)

Numerical results

Table 1 reports the descriptive statistics of the raw DAWS.

Table 1: Descriptive statistics of raw daily average wind speed (DAWS).

Statistic	Value	Notes
n	1,462	daily observations
Mean $\bar{v}$	2.4687 m/s	
Median	2.2417 m/s	< Mean $\Rightarrow$ right skew
Standard deviation	0.9481 m/s	
Variance	0.8988 m ² /s ²	
Minimum	0.9333 m/s	no calm periods in daily average
Maximum	9.1125 m/s	
Skewness	+1.7616	(Normal = 0)
Excess kurtosis	+4.7351	(Normal = 0)
JB statistic	2121.98	
JB p -value	< 0.000001	H_0 (normality) strongly rejected

Interpretation

Every single statistic in Table 1 points in the same direction: the raw DAWS is *fundamentally non-Gaussian*.

Right skewness (+1.76). The mean exceeds the median by about 0.23 m/s, confirming that the distribution has a long right tail. Intuitively, calm days are bounded: the wind cannot blow below zero, and daily averages almost never fall below 0.9 m/s. But occasionally a storm pushes the daily average up to 9 m/s — these rare, high-speed days pull the mean above the median and stretch the right tail. This is the canonical wind speed shape described in Garcia et al. (1998), who find that mean > median holds universally across all 20 of their Navarre stations.

Excess kurtosis (+4.74). A normal distribution has excess kurtosis of exactly zero. A value of +4.74 means the distribution has much heavier tails than a Gaussian — extreme wind speeds (both very low and very high daily averages) occur more frequently than normality would predict. This is a known property of wind speed confirmed in Alexandrinis (2013), who reports strong positive kurtosis for New York DAWS, and in Tol (1997), who motivates ARCH/GARCH modelling of wind precisely because the residuals after a seasonal mean removal remain heavy-tailed.

Jarque-Bera test. The JB statistic combines skewness and kurtosis into a single test of normality:

$$JB = \frac{n}{6} \left[S^2 + \frac{(K)^2}{4} \right] = \frac{1462}{6} \left[1.76^2 + \frac{4.74^2}{4} \right] = 2121.98, \quad (2)$$

where S is skewness and K is excess kurtosis. Under H_0 (normality), this statistic follows a $\chi^2(2)$ distribution. A value of 2122 against a $\chi^2(2)$ critical value of about 6 at the 5% level leaves absolutely no room for doubt: **raw wind speed cannot be modelled as Gaussian**. This result is the motivating justification for everything that follows in Phase 1.

Literature comparison. Alexandrinis (2013, Table 1) reports JB p -values of ≈ 0 for New York DAWS, with skewness of +0.82 and kurtosis of +1.71. Our values are larger in magnitude (skew = +1.76, kurtosis = +4.74), consistent with the lower mean wind speed of our dataset: at 10 m height the distribution is more compressed near zero and therefore more skewed than at the higher-speed regime of New York. Šaltytė Benth & Benth (2008) similarly report strong non-normality and positive skew before the Box-Cox transformation.

Marginal Distribution Fitting (Cells 3 & 7)

Why fit a marginal distribution?

Before fitting any time series model, we must understand the *marginal* (unconditional) shape of the distribution of wind speed. This matters for two reasons. First, the marginal distribution governs the *mean energy production* of a turbine: integrating the turbine's power curve $P(v)$ against the fitted density $f(v)$ gives the expected annual energy production (AEP) — this is the direct link between statistical modelling and the physical asset that a wind derivative is written on. Second, the choice of marginal distribution partly guides the Box-Cox transformation: a Weibull-distributed series requires a different power transformation to achieve normality than a lognormal series.

The Weibull distribution

The Weibull distribution has PDF:

$$f_W(v) = \frac{k}{c} \left(\frac{v}{c}\right)^{k-1} \exp\left[-\left(\frac{v}{c}\right)^k\right], \quad v \geq 0, \quad (3)$$

where $k > 0$ is the shape parameter and $c > 0$ is the scale parameter. The theoretical mean is $\bar{v} = c \Gamma(1 + 1/k)$, where Γ denotes the gamma function. The Weibull's claim to being the natural distribution for wind speed rests on a theorem proved by Tuller & Brett (1984): when the two horizontal components of wind velocity are independently normally distributed with mean zero and equal variance, the marginal distribution of wind *speed* (the magnitude of the vector) is exactly Weibull with $k = 2$ (the Rayleigh special case). More generally, if the two components are not perfectly isotropic but approximately so — which is a reasonable assumption in the free atmosphere above the surface layer — the resulting speed distribution is Weibull with k near 2. This is a physical, not merely empirical, justification.

Parameters were estimated by two methods:

1. **OLS linearisation** (Garcia et al. 1998). The Weibull CDF, $F(v) = 1 - \exp[-(v/c)^k]$, can be double-log-transformed to yield a linear relationship: $\ln[-\ln(1 - F(v))] = k \ln v - k \ln c$. Plotting this against $\ln v$ (“Weibull probability paper”) and fitting by OLS yields estimates $\hat{k}_{\text{OLS}} = 3.2496$ and $\hat{c}_{\text{OLS}} = 3.0699$ m/s.

2. **Maximum likelihood (MLE)**. Direct numerical maximisation of the Weibull log-likelihood gives $\hat{k}_{\text{MLE}} = 2.6368$ and $\hat{c}_{\text{MLE}} = 2.7729$ m/s. The MLE is the statistically efficient estimator and is preferred for the AIC/BIC comparison.

The large discrepancy between the OLS and MLE shape estimates ($k = 3.25$ vs. $k = 2.64$) is itself diagnostic. It arises because the OLS linearisation weights all points in log-probability space equally, meaning that the sparse upper tail (speeds above 5 m/s represent fewer than 10% of observations) exerts disproportionate leverage on the regression line. The MLE, which weights each observation by its actual likelihood contribution, is less sensitive to this sparsity. For all quantitative comparisons we therefore prefer the MLE.

The lognormal distribution

The lognormal distribution has PDF:

$$f_L(v) = \frac{1}{\sqrt{2\pi} s v} \exp\left[-\frac{(\ln v - m)^2}{2s^2}\right], \quad v > 0, \quad (4)$$

where $m = \mathbb{E}[\ln v]$ and $s = \text{Std}[\ln v]$. Parameters are estimated directly from the log-transformed series: $\hat{m} = 0.8423$ and $\hat{s} = 0.3405$.

Distribution comparison

Table 2: AIC/BIC comparison of marginal distribution models. Both models have $k_{\text{params}} = 2$; the decisive quantity is the log-likelihood.

Model	Log-likelihood	AIC	BIC
Weibull (OLS parameters)	−2062.33	4128.66	4139.24
Weibull (MLE parameters)	−1960.51	3925.01	3935.59
Lognormal	−1730.56	3465.12	3475.69

The AIC is defined as $\text{AIC} = -2\ell + 2k$ where ℓ is the log-likelihood and k is the number of free parameters (here $k = 2$ for both distributions, so the AIC difference equals twice the log-likelihood difference). The BIC adds a stronger penalty for model complexity: $\text{BIC} = -2\ell + k \ln n$. Both criteria unambiguously select the lognormal, with $\Delta\text{AIC}(\text{Weibull MLE vs. Lognormal}) \approx 460$ units — a gap that cannot be attributed to sampling variation.

Why does the lognormal win on this dataset?

This result does not contradict the broader literature (where Weibull almost always wins). Several dataset-specific features explain it.

Measurement height. The data are at 10 m, not hub height. Tuller & Brett’s (1984) theoretical derivation of the Weibull assumes the wind is in the free atmosphere where horizontal velocity components are approximately bivariate-normal. This assumption breaks down near the surface, where topographic friction, thermal convection, and daily boundary-layer cycles distort the shape of the speed distribution. The lognormal’s extra

flexibility in the lower tail — it assigns zero probability to $v = 0$ but rises more slowly from zero than the Weibull with $k \approx 2.6$ — better describes this surface-layer regime.

Low mean speed. At $\bar{v} = 2.47$ m/s, the Weibull with $k = 2.64$ and $c = 2.77$ assigns non-negligible probability mass to speeds below the observed minimum of 0.93 m/s. Daily averages never reach near zero because calm hours are always mixed with moving air within the same day. The lognormal, with its lower left tail, fits this empirical absence of very low daily averages more accurately.

Short sample. With only 4 years of data, distributional comparisons are noisy. The 460-unit AIC gap reflects the fit over the 1–5 m/s bulk of the distribution where observations concentrate.

In summary: *the lognormal fits this proxy dataset better for physically interpretable reasons, and the Weibull remains the theoretically preferred and empirically superior choice at hub height, consistent with all cited literature.*

Box-Cox Transformation (Cells 4 & 6)

Why a transformation is necessary

The entire downstream modelling pipeline — seasonal decomposition, AR(4) fitting, CAR(4) embedding, and closed-form futures pricing — rests on the assumption that the residuals of the model are Gaussian. Raw wind speed is manifestly non-Gaussian (Section 3). A transformation that maps the marginal distribution to approximate normality is therefore not merely a convenience: it is a *structural prerequisite*.

The Box-Cox family

The Box-Cox transformation is:

$$W^{(\lambda)}(t) = \begin{cases} \frac{W(t)^\lambda - 1}{\lambda} & \lambda \neq 0, \\ \ln W(t) & \lambda = 0. \end{cases} \quad (5)$$

The parameter λ controls the strength of the power transformation. $\lambda = 1$ is the identity (no transformation). $\lambda = 0$ is the log-transform. Negative λ correspond to reciprocal-type transforms. The optimal λ is estimated by maximising the log-likelihood of the transformed data under a normal distribution assumption, which is equivalent to minimising:

$$\mathcal{L}(\lambda) = \frac{n}{2} \ln \hat{\sigma}^2(\lambda) - (\lambda - 1) \sum_{t=1}^n \ln W(t), \quad (6)$$

where $\hat{\sigma}^2(\lambda)$ is the sample variance of the transformed series. The second term acts as a Jacobian correction.

Results

Table 3: Normality diagnostics across transformations. Ex.Kurt is excess kurtosis (normal = 0). JB and KS are the Jarque-Bera and Kolmogorov-Smirnov p -values (> 0.05 is a pass).

Transformation	Skewness	Ex. Kurtosis	JB p	KS p
Raw ($\lambda = 1$)	+1.762	+4.735	0.000	—
Log-transform ($\lambda = 0$)	+0.487	+0.443	0.000	—
Optimal BC ($\lambda = -0.450$)	-0.005	+0.128	0.606	0.146
Rounded BC ($\lambda = 0.1$)	+0.601	+0.618	0.000	0.000

The optimal $\lambda = -0.450$ achieves essentially perfect Gaussianisation: skewness of -0.005 (compared to the ideal of 0), excess kurtosis of $+0.128$ (compared to the ideal of 0), and a JB p -value of 0.606 — the null hypothesis of normality cannot be rejected at any reasonable significance level. This is exactly the kind of result Šaltytė Benth & Benth (2008) report for New York, where they find $\lambda = 0.2$ achieves approximate normality.

The rounding to $\lambda = 0.1$. Although $\lambda = -0.450$ achieves perfect normality, it cannot be used directly in the pricing formulas. The reason is algebraic: the wind speed $W(t)$ appears in the payoff function as $W(t)^{1/\lambda}$, and the computation of $\mathbb{E}^Q[W(t)^{1/\lambda}]$ under the risk-neutral measure requires evaluating the $(1/\lambda)$ -th moment of a Gaussian random variable. This moment exists in closed form only when $1/\lambda$ is a positive integer. For $\lambda = -0.450$, we get $1/\lambda \approx -2.22$, which is not an integer and would break the pricing formula. The tractable values are $\lambda = 1$ (identity), $\lambda = 0.5$ ($1/\lambda = 2$), $\lambda = 0.25$ ($1/\lambda = 4$), $\lambda = 0.2$ ($1/\lambda = 5$), and $\lambda = 0.1$ ($1/\lambda = 10$).

The nearest tractable value to -0.450 is $\lambda = 0.1$, but this is actually a jump of 0.55 units in λ -space. As Table 3 shows, $\lambda = 0.1$ fails to achieve normality (JB $p = 0$, skewness = 0.601). This is a significant cost of the tractability constraint. There is an explicit modelling trade-off here: we are accepting a residual departure from normality in exchange for having a model that can produce closed-form derivative prices. This trade-off becomes less severe in Phase 2, where the higher mean hub-height wind speeds will require a less extreme λ to Gaussianise the data, making the rounded value closer to the optimal.

Literature comparison. Šaltytė Benth & Benth (2008) find $\lambda = 0.2$ optimal for New York DAWS, with both $\lambda = 0.2$ and $\lambda = 0.1$ described as valid tractable choices. Alexandrinis (2013) finds $\lambda = 0.014 \approx 0$, effectively the log-transform, optimal for their dataset. The fact that our optimal is $\lambda = -0.450$ (a reciprocal-type transform) rather than the positive values found in the literature is a direct consequence of the low mean speed at 10 m height, which produces a more extreme right skew requiring a stronger transformation to correct.

Seasonal Mean $\Lambda(t)$ (Cell 8)

What is the seasonal mean and why model it?

Even after the Box-Cox transformation, the transformed series $W^{(\lambda)}(t)$ is not stationary. If you plot it, you will see that the average level is systematically higher in winter (when Atlantic storms bring strong winds) and lower in summer (when anticyclonic conditions dominate). This deterministic, repeating pattern is called the *seasonal mean*. If we leave it in the data and fit an AR model directly, the AR model will try to “explain” the season — it will fit a slowly varying mean that is actually just calendar effects. This is a mis-specification. We must first remove the seasonal mean so that what remains (the residuals) is genuinely random.

The seasonal mean is modelled by a *truncated Fourier series*:

$$\Lambda(t) = a_0 + \sum_{k=1}^{N_H} \left[a_k \cos\left(\frac{2\pi kt}{365}\right) + b_k \sin\left(\frac{2\pi kt}{365}\right) \right]. \quad (7)$$

The idea is elegant: any smooth periodic function with period 365 days can be approximated arbitrarily well by a sum of cosines and sines at the fundamental annual frequency and its harmonics. Keeping $N_H = 2$ harmonics captures the annual cycle (dominant: one peak per year) and the semi-annual cycle (a weaker pattern that peaks twice per year). Parameters are estimated by ordinary least squares (OLS), which is simply a linear regression of $W^{(\lambda)}(t)$ on the $2N_H + 1 = 5$ basis functions.

Results

Table 4: OLS estimates of the seasonal mean $\Lambda(t)$. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Parameter	Description	Estimate	Significance
a_0	Intercept (mean of transformed series)	0.88513	***
a_1 (cos)	Annual cosine amplitude	−0.00544	—
b_1 (sin)	Annual sine amplitude	0.16381	***
a_2 (cos)	Semi-annual cosine amplitude	0.00887	—
b_2 (sin)	Semi-annual sine amplitude	0.09149	***
$R^2 = 0.1262$, $R_{\text{adj}}^2 = 0.1238$			

Interpretation

Significance pattern. A Fourier harmonic consists of both a cosine and a sine term. Together they describe a sinusoidal wave of a certain amplitude and phase:

$$a_k \cos\left(\frac{2\pi kt}{365}\right) + b_k \sin\left(\frac{2\pi kt}{365}\right) = R_k \cos\left(\frac{2\pi kt}{365} - \phi_k\right), \quad (8)$$

where $R_k = \sqrt{a_k^2 + b_k^2}$ is the amplitude and $\phi_k = \arctan(b_k/a_k)$ is the phase offset. The cosine term a_k sets the amplitude of the component that peaks at $t = 0$ (1 January); the sine term b_k sets the amplitude of the component that is shifted 90 days from 1 January.

The fact that b_1 is significant but a_1 is not means: the annual cycle exists (there is genuine wind seasonality), but its peak does not fall on 1 January — it is phase-shifted into the season by about $\phi_1 = \arctan(0.164 / -0.005) \approx 90$ degrees, placing the maximum in early spring (around late March/April). This is physically sensible.

Low R^2 . The seasonal mean explains only 12.6% of the total variance in the transformed series. This is low compared to what Šaltytė Benth & Benth (2008) find on the New York 20-year dataset, where the seasonal mean is a much more prominent feature. The explanation is twofold: (1) the dataset covers only 4 years, so year-to-year weather variability dominates the seasonal signal; (2) the measurement site may have weaker seasonal structure than a coastal station. A low R^2 for the seasonal mean does not indicate a modelling failure — it simply means that most of the variance in this dataset is driven by day-to-day weather (which the AR model will capture) rather than by calendar seasonality.

Residuals. The mean of $\varepsilon(t) = W^{(\lambda)}(t) - \Lambda(t)$ is -0.000000 (effectively machine zero) and the standard deviation is 0.3494. The near-zero mean confirms that the OLS fit has correctly centred the residuals. The Ljung-Box p -value at lag 5 is 0.000, confirming strong residual autocorrelation — this is *expected* at this stage, and is precisely what the AR(4) model in Phase 2 will remove.

Seasonal Variance $\sigma^2(t)$ (Cell 9)

Why model seasonal variance separately?

After removing the seasonal mean, the residuals $\varepsilon(t)$ should be zero-mean and stationary in *mean* — but they may still be non-stationary in *variance*. Empirically, it is well-established (Tol 1997; Šaltytė Benth & Benth 2008) that wind speed residuals are more volatile in certain seasons (typically winter and early spring, when synoptic weather systems are most active) and calmer in summer. If this pattern is not removed, the standardised residuals will have a variance that varies with the calendar day, violating the i.i.d. assumption required for the CAR(4) model.

The approach follows Šaltytė Benth & Benth (2008) exactly: compute the empirical variance of $\varepsilon(t)$ per calendar day (grouping all available years), then fit a smooth function to these 365 empirical variances using a *cosine-only* Fourier series:

$$\sigma^2(t) = c_0 + \sum_{k=1}^{N_V} c_k \cos\left(\frac{2\pi kt}{365}\right). \quad (9)$$

The cosine-only specification (no sine terms) reflects the approximate symmetry of the variance function around its annual peak: variance rises and falls symmetrically over the year, without a strong phase-shift of the kind captured by the sine terms in the mean.

Results

Table 5: OLS estimates of the seasonal variance $\sigma^2(t)$. *** $p < 0.01$.

Parameter	Description	Estimate	Significance
c_0	Mean variance level	0.13138	***
c_1 (cos)	Annual cosine amplitude	0.09579	***
c_2 (cos)	Semi-annual cosine amplitude	-0.00332	—
$R^2 = 0.2264$; $\sigma^2(t) \in [0.0323, 0.2238]$; $\sigma(t) \in [0.1797, 0.4731]$			

Interpretation

The seasonal pattern. The dominant term is $c_1 = 0.0958$, which has the same sign as $c_0 = 0.1314$. The cosine function $\cos(2\pi t/365)$ equals $+1$ in January and -1 in July, so:

$$\sigma^2(\text{Jan}) \approx 0.1314 + 0.0958 = 0.2272, \quad \sigma^2(\text{Jul}) \approx 0.1314 - 0.0958 = 0.0356. \quad (10)$$

The variance in winter is approximately $6\times$ larger than in summer. This is consistent with the physical picture: winter weather in mid-latitudes is dominated by active frontal systems that produce large day-to-day wind speed swings; summer is dominated by stable anticyclones with little variability. Note that this is the *inverse* of what Šaltytė Benth & Benth (2008) find for the New York dataset, where variance is highest in spring and autumn and lower in winter — a reflection of New York’s different climatological regime. Our dataset’s winter-peak variance is more typical of European Atlantic-facing stations.

Low R^2 . The Fourier fit to the daily variance achieves $R^2 = 0.226$. This is modest, but the code itself provides the correct explanation: with only ≈ 4 observations per calendar day (one per year over 4 years), each per-DOY variance estimate is computed from only 4 numbers. A variance estimated from 4 samples is highly noisy. The low R^2 reflects estimation noise in the target, not a poor Fourier specification. With 20 years of data (as in Šaltytė Benth & Benth 2008), each per-DOY estimate would be based on 20 observations and the R^2 would improve substantially.

Non-negativity enforcement. The code correctly applies $\sigma^2(t) \leftarrow \max(\sigma^2(t), 10^{-8})$ after the Fourier fit. This safeguard prevents numerical issues in the standardisation step: if the Fourier fit were to dip below zero (which can happen with low R^2 and noisy data), dividing by $\sigma(t)$ would produce complex numbers.

Standardised Series $\tilde{W}(t)$ (Cell 10)

Construction

The standardised series is defined as:

$$\tilde{W}(t) = \frac{W^{(\lambda)}(t) - \Lambda(t)}{\sigma(t)} = \frac{\varepsilon(t)}{\sigma(t)}, \quad (11)$$

where $\lambda = 0.1$ is the tractable Box-Cox parameter, $\Lambda(t)$ is the fitted seasonal mean, and $\sigma(t) = \sqrt{\sigma^2(t)}$ is the fitted seasonal standard deviation. The idea is simple: we subtract the expected level for each calendar day, then divide by the expected volatility for that day. If the seasonal mean and variance have been modelled correctly, $\tilde{W}(t)$ should be zero-mean, unit-variance, and have no remaining seasonal structure.

Diagnostic results

Table 6: Diagnostic statistics for the standardised series $\tilde{W}(t)$. Targets are given in parentheses.

Statistic	Value	Status
Mean (target: 0)	0.001244	PASS
Std (target: 1)	0.9667	PASS
Skewness (target: ≈ 0)	+0.5111	Residual (expected)
Pearson kurtosis (target: 3)	3.9253	Residual (expected)
JB p -value (> 0.05)	0.0000	Non-normal (expected)
KS p -value (> 0.05)	0.0595	Borderline
LBQ $\tilde{W}$ at lag 5 (p -value)	0.0000	Autocorr. present
LBQ $\tilde{W}$ at lag 10	0.0000	$\rightarrow$ AR(4) needed
LBQ $\tilde{W}$ at lag 20	0.0000	
LBQ $\tilde{W}^2$ at lag 5	0.0000	GARCH effects
LBQ $\tilde{W}^2$ at lag 10	0.0000	$\rightarrow$ GARCH needed
LBQ $\tilde{W}^2$ at lag 20	0.0000	
$n = 1,462$ observations in <code>W_tilde_series</code>		

Interpretation: what each diagnostic tells us

Mean ≈ 0 and Std ≈ 1 . Both the centring and the scaling have been executed correctly. The mean of $\tilde{W}(t)$ being 0.001244 — effectively machine zero — confirms that the OLS seasonal mean has perfectly centred the series. The standard deviation of 0.9667 is close to 1, indicating that the seasonal variance model has captured most of the heteroskedasticity in the residuals. These are the two most basic sanity checks for the standardisation procedure, and both pass.

Non-zero skewness and kurtosis. The skewness of +0.511 and Pearson kurtosis of 3.925 are not zero, which means $\tilde{W}(t)$ is not yet exactly Gaussian. This is *expected* at this stage and should not be alarming. The reason is that these departures from normality are largely driven by the residual temporal autocorrelation that is still present in $\tilde{W}(t)$. A time series with positive autocorrelation will appear more skewed than its true marginal distribution because consecutive observations are correlated: a high value today tends to be followed by a high value tomorrow, making the joint distribution non-Gaussian even if the marginal is. Once the AR(4) model is applied in Phase 2, the remaining autocorrelation will be removed, and the residuals of the AR(4) are expected to be much closer to Gaussian.

Ljung-Box test on $\tilde{W}(t)$. The Ljung-Box test checks the joint hypothesis that the first m autocorrelations of $\tilde{W}(t)$ are all zero:

$$\text{LBQ}(m) = n(n+2) \sum_{k=1}^m \frac{\hat{\rho}^2(k)}{n-k} \sim \chi^2(m) \quad \text{under } H_0. \quad (12)$$

The p -values are essentially zero at all tested lags (5, 10, 20), confirming that $\tilde{W}(t)$ contains

strong serial autocorrelation. Again, this is the *expected* result at the end of Phase 1: the seasonal mean and variance have been removed, but the short-term weather dynamics (the “memory” of the atmosphere, which means that a windy day tends to be followed by a windy day) have not yet been modelled. The AR(4) in Phase 2 is precisely designed to capture this memory. This finding directly motivates the transition to Phase 2.

Ljung-Box test on $\tilde{W}^2(t)$. The LBQ applied to the *squared* standardised residuals tests for *conditional heteroskedasticity*, which is the phenomenon where the *variance* of the innovation changes over time in a clustering pattern — volatile periods tend to cluster together. The p -values of 0.0000 at all lags confirm that GARCH effects are present in this dataset even after removing the deterministic seasonal variance. This means there is a stochastic, clustering component of volatility on top of the deterministic seasonal pattern modelled by $\sigma^2(t)$. This directly motivates the GARCH extension in Phase 3, which will model this additional layer of variance dynamics. This result is fully consistent with Tol (1997), who is the first paper to document ARCH effects in daily wind speed residuals and argue for their inclusion in wind speed models.

Literature comparison. Šaltytė Benth & Benth (2008) report that after their full preprocessing chain (Box-Cox + Fourier mean + Fourier variance), the standardised residuals have skewness = -0.09 , kurtosis = 3.03 , JB $p = 1$, and LBQ $p = 0.989$. Our $\tilde{W}(t)$ series is at an earlier stage of this chain — we have applied the Box-Cox transformation, seasonal mean, and seasonal variance, but not yet the AR(4) — so the remaining autocorrelation and non-normality in our diagnostics is entirely consistent with theirs. Their fully standardised residuals correspond to what we will compute at the *end* of Phase 2, not Phase 1.

Figure Analysis (Cells 5 and the Phase 1 Plot Set)

Figure 1 — Raw Wind Speed: Statistical Properties (7 panels)

Panel 1 (top, full width): Raw daily wind speed time series. The time series shows the typical structure of daily wind speed: spiky, volatile, with no visible trend but a loose seasonal envelope visible to the eye — slightly higher activity in the winter months of each year. The red dashed line at $\bar{v} = 2.47$ m/s provides the reference level. The irregular spikes (days when the daily average approaches 9 m/s) confirm the heavy-tailed nature documented in the descriptive statistics. The series also appears stationary in mean over the 4-year window, which is encouraging — there is no visible upward or downward trend.

Panel 2 (row 2, left): Raw wind speed histogram with Weibull and lognormal overlays. The histogram shows the characteristic right-skewed, unimodal shape of daily wind speed. The mode (peak of the histogram) is below the mean, and the distribution has a long right tail. Both the Weibull (red) and lognormal (green dashed) curves are visually close to the data. The lognormal appears to track the histogram body slightly better, particularly in the 1–4 m/s range where most observations fall — consistent with the AIC result. The Weibull, with its slightly heavier lower tail, assigns too much density below 1 m/s where the data are sparse.

Panel 3 (row 2, centre): Weibull probability paper. This is the diagnostic plot for the OLS linearisation of the Weibull CDF. If the data were perfectly Weibull, all points would lie exactly on a straight line. The $R^2 = 0.869$ (in probability-paper space) indicates

approximate linearity but with visible deviation in the upper tail, where the data points bend away from the OLS line. This curvature in the upper tail is the signature of the lower k value found by MLE ($k = 2.64$) vs. OLS ($k = 3.25$): the MLE correctly identifies a lighter shape while the OLS line is pulled by the dense central range.

Panel 4 (row 2, right): Q-Q plot — raw wind speed vs. normal. The Q-Q plot compares quantiles of the data against quantiles of a normal distribution. For a Gaussian variable, all points lie on the diagonal. The characteristic “S-curve” deviation here — points above the line at both tails — is the visual signature of positive excess kurtosis (fat tails). The right tail extends far above the normal line, confirming the heavy right tail and strong positive skewness. This plot is perhaps the clearest visual proof that raw wind speed is not Gaussian and that a transformation is required.

Panel 5 (row 3, left): Histogram after optimal Box-Cox ($\lambda = -0.450$). After applying the optimal transformation, the histogram is visually symmetric and the normal curve (black line) passes through the histogram bars accurately. This is the visual confirmation of the JB $p = 0.606$ result: the optimal Box-Cox transformation successfully Gaussianises the distribution. The transformed values range roughly from -4 to $+1$ in transformed units.

Panel 6 (row 3, centre): Histogram after rounded Box-Cox ($\lambda = 0.1$). With the tractable $\lambda = 0.1$, the histogram is less symmetric: a visible right skew remains (the right tail extends further than the normal curve). This is the visual confirmation of skewness $= 0.601$ and JB $p \approx 0$ with the rounded λ . The normal curve does not fit as well as in Panel 5 — the cost of the tractability constraint is visible.

Panel 7 (row 3, right): Q-Q plot after rounded Box-Cox ($\lambda = 0.1$). The Q-Q plot confirms the histogram diagnosis. Compared to Panel 4 (raw), the S-curve deviation is substantially reduced — transformation has clearly improved normality. But the right tail still lifts above the diagonal, confirming residual positive skew. The improvement is real but incomplete, which is consistent with the summary table (Table 3).

Summary and Phase 1 Outcome

Phase 1 has accomplished its defined purpose. Starting from raw hourly wind speed observations, the pipeline has:

1. Documented the non-Gaussian, right-skewed, fat-tailed nature of raw DAWS, quantified by JB $p < 10^{-6}$, skewness $= +1.76$, and excess kurtosis $= +4.73$.
2. Identified that the lognormal distribution provides a better marginal fit than the Weibull on this 10 m proxy dataset ($\Delta\text{AIC} \approx 460$), and explained why this is consistent with the literature’s general preference for Weibull at hub height.
3. Applied the Box-Cox transformation with $\lambda = 0.1$ (tractable choice consistent with Šaltytė Benth & Benth 2008), achieving substantial though incomplete Gaussianisation. The complete normalisation is achieved by the AR(4) fit in Phase 2.
4. Modelled and removed the seasonal mean $\Lambda(t)$ via a 2-harmonic Fourier series (OLS), reducing variance to residuals with $\bar{\varepsilon} \approx 0$ and $\text{Std}(\varepsilon) = 0.3494$.
5. Modelled and removed the seasonal variance $\sigma^2(t)$ via a cosine-only 2-harmonic

Fourier series, producing the standardised series $\tilde{W}(t)$ with mean = 0.001 and std = 0.967.

The series $\tilde{W}(t)$ — stored as `W_tilde_series` — is ready for Phase 2. The LBQ diagnostics confirm: (a) serial autocorrelation is present, motivating AR(4); (b) conditional heteroskedasticity is present, motivating GARCH. Both findings are consistent with the literature and with the modelling roadmap.

Key numbers.

- $n = 1,462$ daily observations, 2015–2018, $\bar{v} = 2.47$ m/s at 10 m height.
- Raw: skewness +1.76, kurtosis +4.74, JB $p < 10^{-6}$.
- Box-Cox $\lambda_{\text{opt}} = -0.45$ (Gaussian), $\lambda = 0.1$ (tractable; $1/\lambda = 10$).
- Seasonal mean $R^2 = 0.126$ (4-year dataset; low, expected).
- Seasonal variance: winter $\hat{\sigma}^2 \approx 0.22$, summer $\hat{\sigma}^2 \approx 0.04$.
- $\tilde{W}(t)$: mean = 0.001, std = 0.967, LBQ $p = 0 \Rightarrow$ AR(4) needed. LBQ on $\tilde{W}^2$ $p = 0 \Rightarrow$ GARCH needed.